Point the Array, Not the Denoiser: Directional Noise Cancellation before DeepFilterNet3 on a 4-Mic ReSpeaker, Judged by Whisper

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Abstract

Service robots often use a 4-microphone array as one microphone. They take the chip's processed channel and clean it with a neural denoiser whose only control is an attenuation limit. We measure, on real hardware, what the four raw microphones of a ReSpeaker USB array add in front of DeepFilterNet3, and how that limit should be set when the listener is a speech recognizer.

One 40-minute session gives 320 scenes. We recorded a talker at 8 positions and 5 interferers from 2 loudspeakers separately on the fixed array, then mixed them, so the speech and noise components of every output are known. Two live takes check the result against reality.

The intuitive fix, switching off the microphone nearest the noise, gains 0.4 dB, because on a 6.4 cm array every microphone hears the noise at the same level. An MVDR beamformer steered by a relative transfer function estimated from the 1.5 s before the talker speaks gains 4.3 dB (oracle: 9.1 dB). At 0 dB input SNR it lowers Whisper's word error rate from 74 % to 50 % ($p = 5\times10^{-9}$; clean speech: 15 %).

DeepFilterNet3 never lowers the word error rate. At the robot's setting of 12 dB the change is within noise (+0.8 points). Without a limit it adds +18.5 points, while ESTOI rises monotonically. Running it in the robot's independent 64 ms chunks adds +11.4 points. Live, with a second talker, the chip's processed channel lets Whisper transcribe the *other* voice (WER 200 %), while MVDR followed by DeepFilterNet3 gets 11 %.

1 Introduction

A typical robot speech pipeline reads one channel from a USB microphone array and denoises it with a neural model. It passes the result to a recognizer. Ours reads the processed channel of a ReSpeaker array, DeepFilterNet3 [14] cleans it with an attenuation limit of 12 dB, and Whisper [12] transcribes it. The other four channels, the raw microphones, are thrown away. Two questions follow:

1. What do those microphones add when the noise comes from a direction?
2. How should the denoiser's only knob, `atten_lim_db`, be set when the listener is a recognizer rather than a person?

A first intuition is to switch off the microphone that faces the noise. It is worth testing because it is how one would reason about directional microphones. The array's microphones are omnidirectional and 4.5 cm apart, so the expectation is that it cannot work. The tool that should work is a beamformer that combines all four with phase. Whether it works on this array, in a real room, with noise statistics a robot can actually obtain, is an empirical question. Project 15, our earlier work on this array, answered it only in simulation.

Contributions.

1. **A recording protocol with exact scoring.** Chirp-aligned speech-only and noise-only takes on a fixed array give exact speech and noise components for every linear front end, and real recordings underneath. Live takes check the protocol.
2. **Hardware numbers for nine front ends** on 320 scenes, from the intuitive mute to an RTF-steered MVDR and an adaptive GSC. Each uses only the 1.5 s before the talker speaks.
3. **A recognition sweep** of DeepFilterNet3's attenuation limit, continuous and in the robot's chunked mode, with 4,016 Whisper transcriptions.
4. **A live comparison with the array's own processed channel.**

2 Related work

Beamforming. The MVDR beamformer [3] and its adaptive form, the generalized sidelobe canceller [6],

are the classic linear tools for directional noise suppression [2, 5]. In reverberant rooms the free-field steering vector is replaced by the relative transfer function (RTF), which can be estimated without geometry by covariance whitening, the principal generalized eigenvector of the noisy and noise-only covariances [10]. Small arrays are the hard case: at 1 kHz a 6.4 cm aperture is a fifth of a wavelength, so any gain beyond delay-and-sum comes from superdirective weights, which are sensitive to steering errors.

Beamforming for ASR. The CHiME challenges [1] established that beamforming before recognition helps, and mask-based covariance estimation [7] made the noise statistics learnable. Here the noise statistics come from the simplest source a robot already has: the seconds before the person speaks, the *silence model* of our earlier work on the same array.

Neural single-channel enhancement. DeepFilterNet [13, 14] is a real-time full-band enhancer that removes stationary and many non-stationary noises. It is trained to sound clean, not to recognize well, and enhancement artifacts are known to hurt ASR even when perceptual metrics improve [8]. Whisper [12] is itself trained on noisy data, which leaves less for a front end to add.

Gap. We found no hardware study of the pipeline as robots actually deploy it: a commodity 4-mic array, a neural denoiser with a user-set attenuation limit, and an off-the-shelf recognizer. Such a study would need both exact signal-level accounting, where the speech and noise components are known, and recognition on live audio.

3 Setup and methods

Hardware. A ReSpeaker USB Mic Array v2.0 with its 6-channel firmware: ch0 is the chip's own processed beam (the channel the robot pipeline uses today), ch1–4 are the raw microphones on a 32 mm circle. An Akane Omnia 300 USB microphone 10–20 cm from the talker's mouth is the close-talk reference. The interferer is a stereo pair of desktop speakers about 0.5 m from the array, driven one side at a time. Seen from the array, the two sides are about 110° apart. On Windows the 4 raw channels are only delivered with the operating system's *audio enhancements* disabled: with them on, every audio API returns 2 identical channels at 48 kHz.

Session. One talker (the author) recorded 40 minutes:

- 15 s of room silence;
- 10 noise-only takes: 5 interferers × 2 speakers, 20 s each;
- 8 speech-only takes: one Spanish prompt read from each of 8 positions around the array at 0.5 m;
- 2 live takes, reading while an interferer plays;
- 7 distance clips for the localization study.

The interferers are pink noise, a synthetic fan, real cafeteria babble (DEMAND [15]), an English talker (LibriSpeech [11]) and synthetic music. Every take starts and ends with a chirp from the speaker. A matched filter on the two chirps aligns each recording to the played signal, and the two USB clocks to each other, to a fraction of a sample. All signals are high-passed at 100 Hz: the raw microphones carry mains hum that dominates their noise power.

Scenes. Speech and noise recorded separately on a fixed array add: $\mathbf{y} = \mathbf{s} + g\,\mathbf{n}$ is what the array would have heard with both sources on. g sets the input SNR to −5, 0, 5 or 10 dB, giving 320 scenes. Each scene begins with 1.5 s in which the talker is silent. That stretch is the only noise information any method may use (the *silence model*). Because every front end below is linear, $\mathbf{s}$ and $\mathbf{n}$ pass through the same weights separately, which makes the output SNR exact.

Front ends (STFT 512/128 at 16 kHz, output referenced to ch1):

- **raw mic**: ch1 alone.
- **mute**: the intuitive idea of switching off the microphone that hears the noise most. We drop the mic with the largest silence-model power and average the rest. **soft** is its per-band version: real inverse-noise-power weights.
- **delay-and-sum** and **MVDR (DOA)**: far-field steering at the direction estimated by SRP-PHAT [4] on the mixture minus the silence model.
- **MVDR (RTF)**: steering by the RTF from covariance whitening [10], with no geometry and no DOA.
- **GSC**: a fixed beam along the RTF, a blocking matrix orthogonal to it, and a per-bin NLMS canceller [6] that adapts only while the fixed beam stays within 3 dB of its noise-only level.
- **oracle**: MVDR with the true noise covariance and speech RTF, as an upper bound.

DeepFilterNet3 and recognition. Each output goes through DeepFilterNet3 [14] with `atten_lim_db` $\in \{6, 12, 18, 24, \infty\}$. The value 12 is the robot's setting. A value of 0 also means *no limit* in the library, not off.

- **Continuous mode** runs one pass per utterance.

- **Chunk mode** reproduces the robot pipeline: 64 ms chunks, each peak-normalized and enhanced independently.

Whisper `small` [12] transcribes every version. Its configuration is greedy decoding, Spanish, no VAD filter and no context. WER is computed against the prompt. ESTOI [9] is computed against the close-talk microphone. The 0 dB scenes get the full sweep.

4 Results

4.1 Signal level: what each front end removes

Figure 1a ranks the front ends by exact SNR gain.

Muting cannot work. Switching off the noisiest microphone gains 0.4 dB and its per-band version 0.5 dB, no more than simply averaging the four (0.4 dB). A far-field source reaches four omnidirectional microphones within 6.4 cm at the same level, so there is no "noisier" microphone to drop. Delay-and-sum also gives 0.4 dB: at speech frequencies the aperture is a fraction of a wavelength, so phase alignment alone builds almost no directivity.

MVDR works. The front ends that whiten the noise do:

- MVDR with an RTF from covariance whitening: 4.3 dB;
- the adaptive GSC: 3.7 dB;
- MVDR steered by the estimated direction: 2.9 dB.

The oracle MVDR, with the true noise covariance and speech RTF, reaches 9.1 dB. That leaves about 5 dB between what the silence model supports and what the array could do. The gain holds across separations (Fig. 1b), and is lowest when talker and interferer are within 45° of each other.

Why the DOA-steered MVDR lags. Its far-field steering depends on a direction estimate (Fig. 2). With an interferer at 0 dB, the direction from the mixture minus the silence model is off by a median 58°. At 10 dB the error falls to 5°. Steering by the RTF avoids the question: it learns the talker's real transfer function, reverberation and near-field included.

4.2 Recognition: the attenuation limit

Figure 3 shows the central result.

The front end dominates. Without DeepFilterNet3, Whisper's WER is 74 % on the raw microphone, 73 % with the mute, 54 % with the GSC and 50 % with the RTF MVDR. The paired difference against the raw microphone is -24.4 points ($p = 5\times10^{-9}$, Wilcoxon over scenes). With the oracle the WER is 20 %, close to the clean floor of 15 %.

DeepFilterNet3 does not help the recognizer. Its effect is the same on every front end:

- At 6 and 12 dB the WER barely moves: +2.7 and +0.8 points at 12 dB, $p = 0.46$ and 0.48.
- From 18 dB on, the WER rises: without a limit, by +16.1 and +18.5 points.
- Meanwhile ESTOI rises monotonically with the limit (Fig. 3b).

This is the perceptual-versus-recognition split known from enhancement artifacts [8]. For Whisper, the best limit is "off" to 12 dB, and the robot's 12 dB is at the safe edge.

The robot's chunked mode costs accuracy. Enhancing each 64 ms chunk independently, peak-normalized, adds +11.4 points after MVDR ($p = 1\times10^{-9}$) and +6.3 on the raw microphone ($p = 3\times10^{-5}$), relative to one continuous pass at the same limit. The recurrent state of DeepFilterNet3 never warms up, and each chunk is rescaled on its own.

By interferer (Fig. 4):

- The spatial front end helps most against the second talker (97 % → 60 %) and the cafeteria (84 % → 52 %), the two interferers that are speech.
- DeepFilterNet3 does not help even for stationary pink noise: on the raw microphone it raises the WER from 78 % to 94 %.
- Music is the easiest interferer at equal SNR.

4.3 Live takes

Superposition check. The talker live take replayed the same interferer samples as its noise-only take. Before the talker speaks, the live recording matches the scaled noise-only take with a residual of -6.4 dB. Most of the residual sits below 300 Hz, in room noise that differs between takes. The rest comes from the talker's body and the changed loudspeaker volume. Subtracting the matched interferer gives the live speech component, and on it the RTF MVDR gains 2.9 dB (DOA-steered: 3.3 dB; mute: -0.5 dB). These are somewhat below the mixed scenes, as the residual suggests.

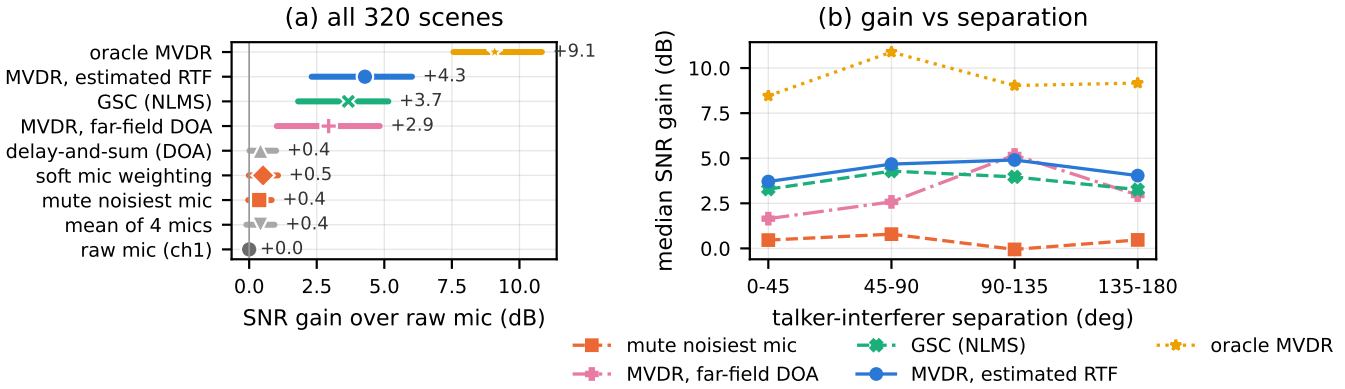


Figure 1: Exact SNR gain over the raw microphone (output minus input SNR on ch1, the same weights applied to the true speech and noise components). (a) Median and interquartile range over all 320 scenes. (b) Median gain against the angle between talker and interferer as seen by the array, each measured from its own clean take.

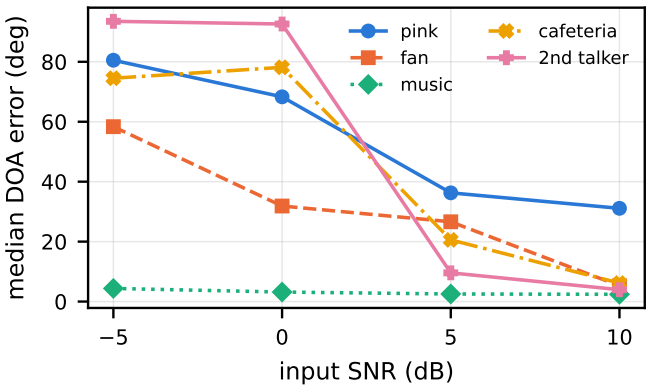


Figure 2: Talker direction estimated from the mixture (SRP-PHAT on mixture minus silence model) against the direction from the clean take.

Live recognition (Fig. 5). With an English talker playing:

- On the chip's processed channel, with or without DeepFilterNet3, Whisper produces a Spanish rendering of the *interfering* voice: WER 200 %. The chip's own beam follows the louder voice.
- The raw microphone with DeepFilterNet3 reaches 44 %.
- Every phase-aware front end gets 7–15 %. The RTF MVDR followed by DeepFilterNet3 gets 11 %.

In the cafeteria take the beamformers reach 16 %, against 48 % for the chip channel.

5 Discussion

Hypotheses.

- **H1, holds.** Muting or weighting microphones gains under 1 dB (0.4 and 0.5 dB, WER -0.8 points, $p = 0.47$). Phase-aware front ends gain 3–4 dB.
  - We predicted *at least 5 dB* for MVDR. Only the oracle reaches that, so the bottleneck is the noise model from 1.5 s of silence, not the array.
- **H2, half right.** The spatial front end matters most for speech-like interferers, as predicted. But DeepFilterNet3 alone does not "handle" stationary noise for the recognizer either: it improves ESTOI and leaves the WER flat or worse.
- **H3, holds, but the optimum does not move.** WER is minimal from "off" to 12 dB and grows beyond, both with and without a beamformer in front.
- **H4, holds.** The robot's independent 64 ms chunks cost 6–11 points.

Falsification test. We set this test in advance: MVDR→DeepFilterNet3 against DeepFilterNet3 alone at noise angles $\geq 90°$. On those 44 scenes the difference is −26.3 WER points, 38 % relative, far beyond the 10 % threshold. The difference is the same below 90°, and over all 0 dB scenes it is -26.3 points ($p = 1 \times 10^{-8}$). The 4-microphone front end is worth adding.

Recommendations for the robot.

1. Read the raw channels, not ch0.
2. Run an RTF-steered MVDR whose noise covariance is updated in the silence before each utterance. The voice activity detector already provides that silence.
3. Run DeepFilterNet3 continuously, or not at all, at 12 dB or below. Never run it without a limit in front of the recognizer.

The setting `atten_lim_db` = 0 deserves a warning, because the library reads it as "no limit".

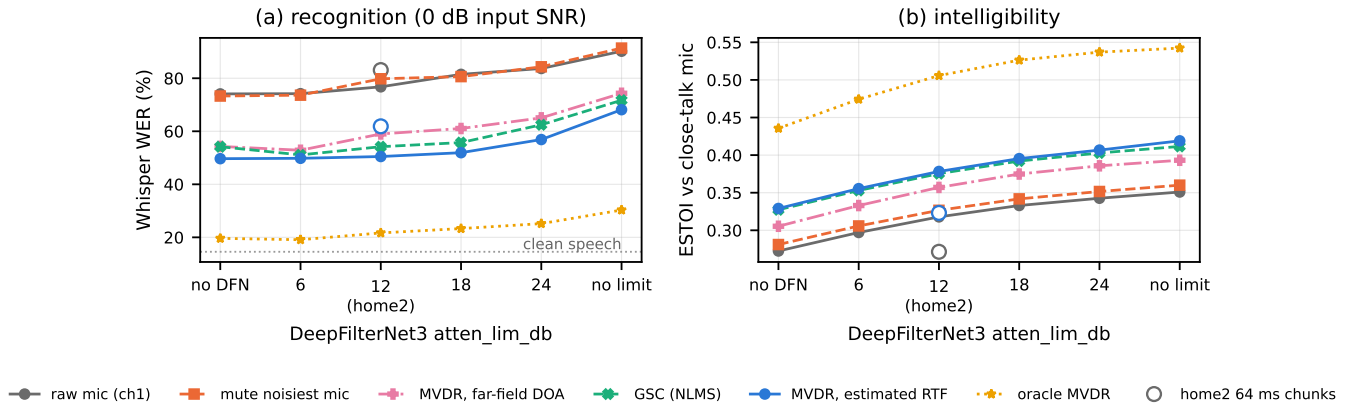


Figure 3: All 0 dB scenes (80), each front end followed by DeepFilterNet3 at every attenuation limit. (a) Whisper WER against the read prompt; dotted: the same Whisper on the clean speech-only take. (b) ESTOI against the close-talk microphone. Hollow markers: the robot's mode, with independent 64 ms chunks, at 12 dB.

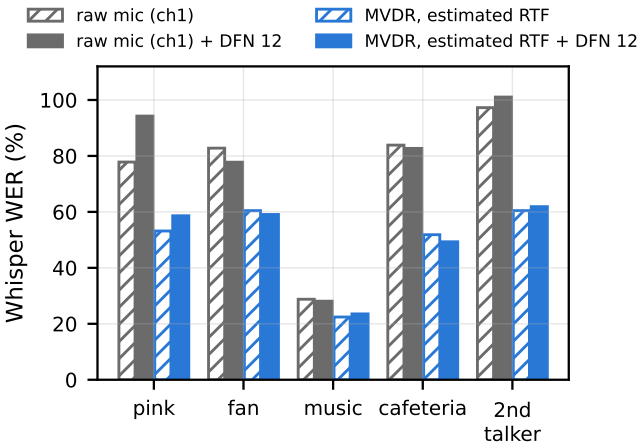


Figure 4: WER by interferer at 0 dB SNR, raw microphone against RTF MVDR, each without (hatched) and with DeepFilterNet3 at 12 dB (solid).

Limitations.

- One talker, one room and one session. The talker's angle marks were approximate, so separations come from the clean takes.
- Scenes are mixed. The live check shows they are optimistic by 1–2 dB of SNR gain, but the two live takes agree with them in direction and size.
- Whisper `small` with greedy decoding; a larger model is more noise-robust and may shrink every gap.
- The 0 dB scenes are hard on purpose: the absolute WERs are high, the differences are what matters.
- The cafeteria live take could not be checked for superposition: a bug in the noise generator replayed a different excerpt.

Project 15. The same session gives project 15 its missing hardware numbers:

- **Orientation.** The array is mirrored and rotated by 35° relative to the talker's marks.
- **Direction.** The leave-one-out median error is 16°, bounded by the marks' accuracy.
- **Distance fails.** No distance cue correlates with distance ($|\text{Spearman}| \leq 0.30$). The best model's error is 0.40 m, against 0.41 m for predicting the mean.

6 Conclusion

On a 4-microphone ReSpeaker, directional noise cancellation has to come from combining the microphones with phase. Switching one off does nothing. An MVDR beamformer steered by a transfer function learned in the silence before the talker speaks cuts Whisper's word error rate by a third at 0 dB SNR. Against a second voice, it is the difference between transcribing the talker and transcribing the television.

DeepFilterNet3 does not add to recognition at any attenuation limit. Above 12 dB, and in the robot's chunked mode, it subtracts, even as intelligibility metrics improve. The limit should be set for the recognizer, not by ear, and the array should be pointed before the denoiser is turned up.

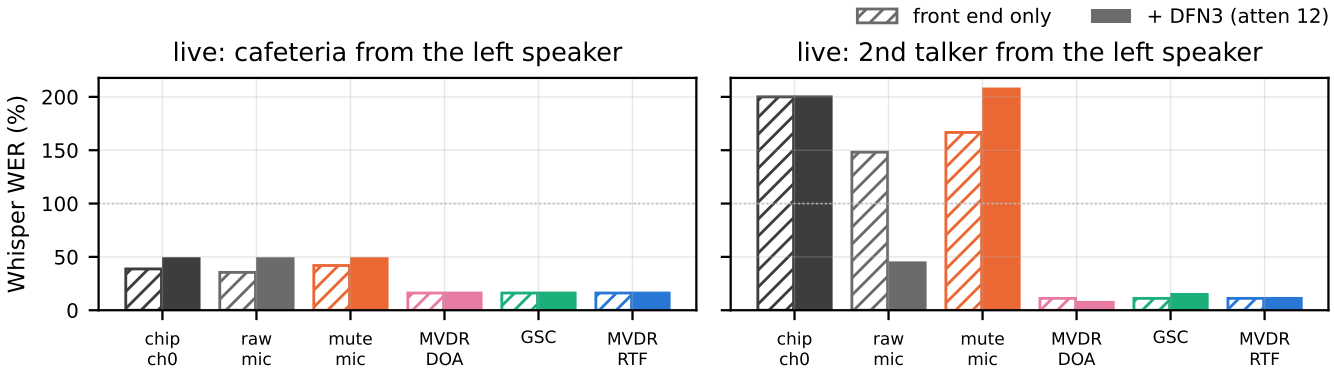


Figure 5: Live takes: the talker reads while an interferer plays from the left speaker. Hatched: front end only. Solid: followed by DeepFilterNet3 at 12 dB. "chip ch0" is the array's own processed channel, the robot's current input. WER above 100 % means Whisper transcribed the interfering voice.

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